Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course: B.Tech
Code: 101405

Subject: Introduction to Solid Mechanics

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) What is the relationship between Young's modulus (E), Bulk modulus (K), and Poisson's ratio (v)?
(i) E=3K(1−2v)
(ii) E=2K(1+v)
(iii) E=K(1+v)
(iv) E=3K(1+v)

(b) Which law governs the linear relationship between stress and strain in the elastic region?
(i) Newton's Law
(ii) Hooke's Law
(iii) Pascal's Law
(iv) Bernoulli's Principle

(c) What is the point on a beam where the bending moment changes sign called?
(i) Neutral Axis
(ii) Point of Contra-flexure
(iii) Point of Zero Shear
(iv) Elastic Center

(d) Which type of loading results in the maximum strain energy stored in a material?
(i) Gradual Loading
(ii) Uniform Loading
(iii) Impact Loading
(iv) Sudden Loading

(e) What is the shear stress distribution across a circular section?
(i) Uniform throughout
(ii) Zero at the center and maximum at the edges
(iii) Maximum at the center and zero at the edges
(iv) Linear across the section

(f) In the bending stress formula $\sigma = \frac{M}{I}y$, what does 'I' represent?
(i) Moment of Inertia
(ii) Section Modulus
(iii) Bending Moment
(iv) Radius of Curvature

(g) What is the slope at the free end of a cantilever beam subjected to a point load at the free end?
(i) Zero
(ii) Uniform
(iii) Maximum
(iv) Undefined

(h) In torsion, what happens to the shear stress as the radius increases?
(i) Decreases
(ii) Increases Linearly
(iii) Remains Constant
(iv) Increases Exponentially

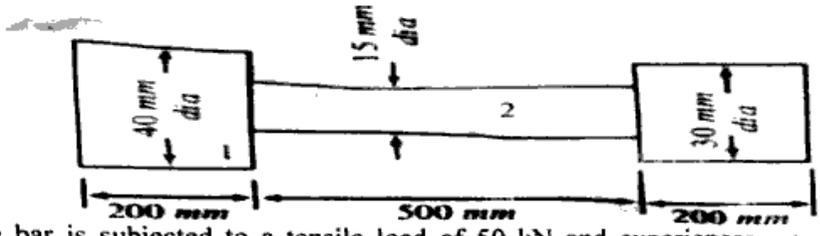
(i) Which parameter affects the torsional rigidity of a shaft?
(i) Length of the shaft
(ii) Density of the shaft materia
(iii) Radius of the shaft
(iv) Temperature of the shaft

(j) What is the longitudinal stress in a thin cylinder subjected to internal pressure?
(i) $\frac{pd}{4t}$
(ii) $\frac{pd}{t}$
(iii) $\frac{pd}{2t}$
(iv) $\frac{p}{t}$

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Q.2 (a) Define stress and strain. Explain the significance of St. Venant's principle in stress analysis. [7]

(b) A steel bar consists of three segments with different cross-sections: [7]
Segment 1: 200 mm long, 40 mm diameter, Segment 2: 500 mm long, 15 mm diameter, Segment 3: 200 mm long, 30 mm diameter.



The bar is subjected to a tensile load of 50 kN and experiences a temperature increase of 30°C. Given: E (steel) = 200 GPa. and α (Coefficient of thermal expansion) = 12 × 10^{-6} /°C. Find the total extension of the bar due to both mechanical load and temperature rise.

Q.3 (a) Explain the concept of lateral strain and derive an expression for Poisson's ratio. [7]

(b) Discuss the significance of section modulus and calculate it for a rectangular and circular section [7]

Q.4 (a) Explain the concept of resilience and derive expressions for strain energy under gradual and sudden loading. [7]

(b) A simply supported beam of length 6 m carries a point load of 10 kN at a distance of 2 m from the left support, a UDL of 2 kN/m upto a distance of 2 m from the right support and a clockwise moment of 5 kN-m at the centre of beam. [7]
i. Draw the Shear Force (SF) and Bending Moment (BM) diagrams.
ii. Find the Maximum Bending Moment and its position.

Q.5 (a) Explain Mohr's circle for stress and use it to find the principal stresses in a 2D stress system. [7]

(b) A structural member is subjected to Normal stress σ_x = 120 MPa (Tensile), σ_y = 50 MPa (Compressive) and Shear stress τ_{xy} = 40 MPa. Draw Mohr's circle and determine: [7]
(i) Principal stresses and their directions. (ii) Maximum shear stress and its plane.

Q.6 (a) Derive the relation between Young's modulus (E), Bulk modulus (K), and Poisson's ratio (ν). [7]

(b) Derive the formula for hoop stress and longitudinal stress in thin-walled cylindrical shells under internal pressure. [7]

Q.7 (a) Construct the shear force and bending moment diagrams for a simply supported beam with a uniformly distributed load (UDL). [7]

(b) A thin cylindrical vessel of 2 m length and 500 mm internal diameter holds a fluid under a pressure of 4 MPa. The thickness of the shell is 10 mm. Calculate: [7]
(i) Hoop stress (circumferential stress) (ii) Longitudinal stress
(iii) Volumetric strain if E = 200 GPa and Poisson's ratio = 0.3

Q.8 (a) Explain the failure theories for materials under combined bending and torsion. [7]

(b) A hollow shaft has an external diameter of 100 mm and an internal diameter of 60 mm. It transmits 120 kW of power at 300 rpm. Calculate: [7]
i. The maximum shear stress in the shaft.
ii. The angle of twist per meter length if G = 80 GPa.

Q.9 Write Short notes on ***any two*** of the following:- [7 x 2 = 14]
(a) Stress-strain diagram for mild steel, including key points.
(b) Hooke's law and its significance.
(c) Euler's Buckling Theory for both end pinned Column.
(d) Torsion of Circular Shaft and Torsion Formula.

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